CS102.3 - Programming with C

Group Assignment – Group No. 10

Bank Management System

Submission deadline: 20th February 2024

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1.Introduction:

The Bank Management System is an application for maintaining a person's account in a bank. The system provides the access to the costumer to create an account, deposit/withdraw the cash from his account, also to convert currency. The following documentation provides the specification of the system.

We are mainly concerned with developing a banking system where a Customer can submit his/her deposit amount to bank if he/she has an account or can create a new account in this bank. Customer can also view the status and change currency of his/her bank account, can view account balance. One can easily maintain the above things if he/she has an account by login through his unique account number

What we are built:-

In this project, we will build a small banking management system in C language.In which we can add accounts,search transactions and many more.

2.Project requirments:

2.1: ER Diagram :-

*Creating a detailed Entity-Relationship (ER) diagram for a bank management system involves identifying and representing various entities, their attributes, and the relationships between them. Below is a simplified ER diagram for a basic bank management system:

Entities:- 1.account

2.Transaction

3.Account holder

Attributes:- 1.Acc_balance

2.Acc_number

3.Holders_name

4.Transaction ID

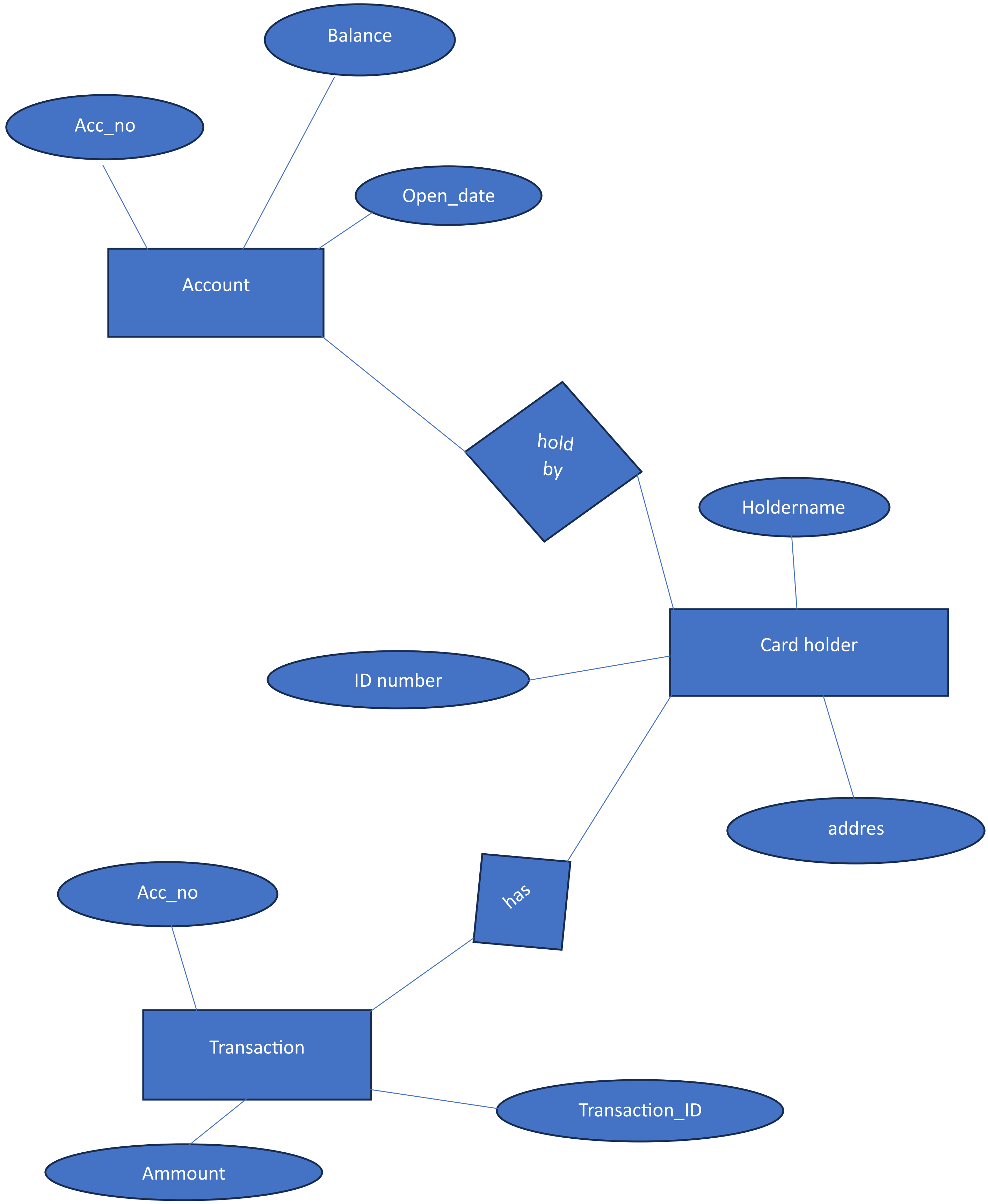
5.Deposite amount

6.Withdrawal amount

Relations:- 1.hold by

2.has

Structure Of ER Diagram



2.2.Data Base

Pre requiments:-

- Header files
- Increment operator
- Functions
- Switch cases
- Basic understanding pf arrays

Header files:- Declare the function prototypes for functions defined in other source file

#include<stdio.h>

#include<stdlib.h>

Increment operator:- basically used to keep track of count **(i ++)**

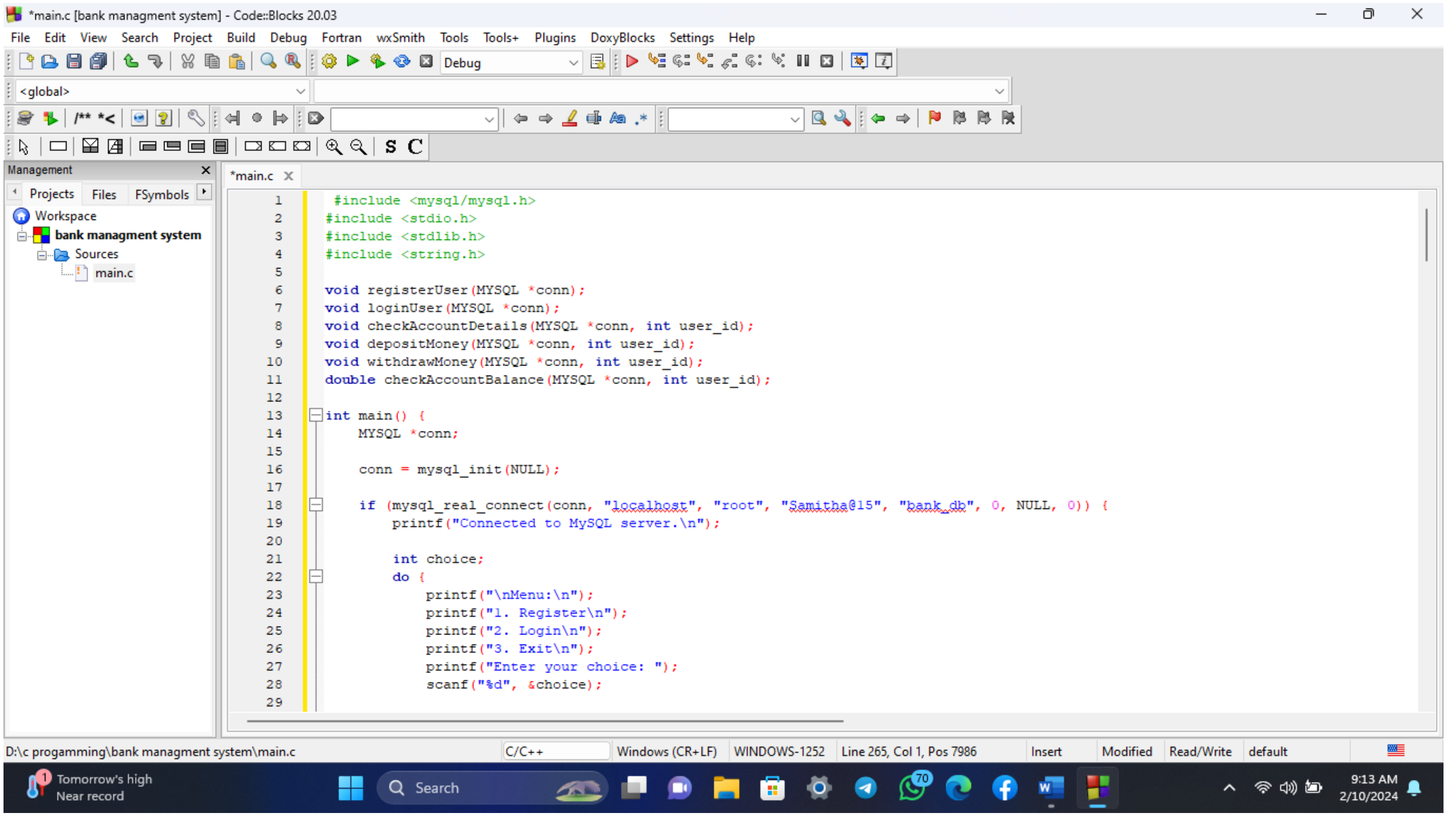
Function:- A block of code which only runs when it is called

Switch cases:-evaluates a given expression and based on the evaluated value, it executes the statements associated with it.

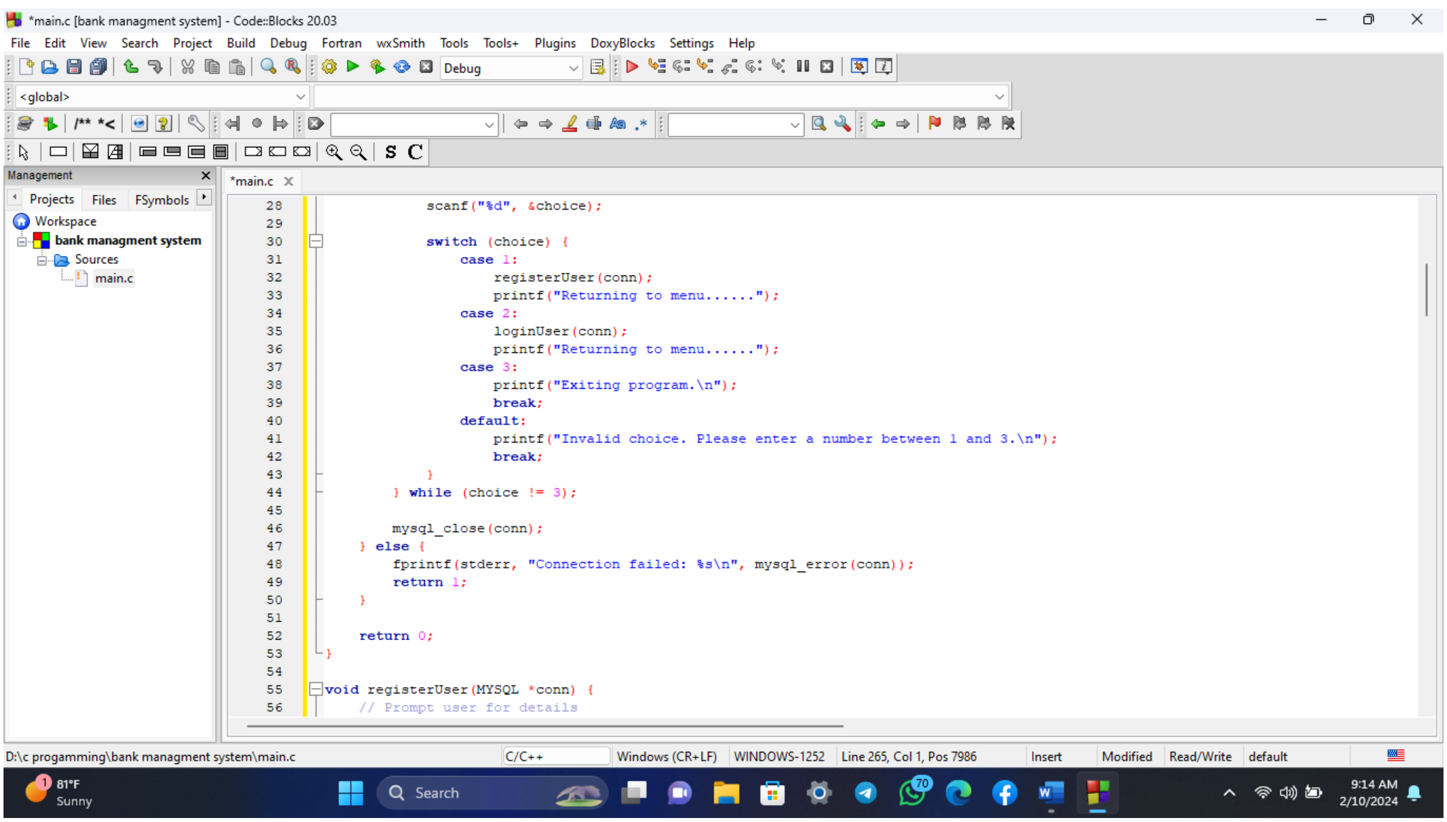
How are we going to build the database system:-

Our strategy will be to develop distinct functions for each activity which will then be combined to from the project's functional unit.

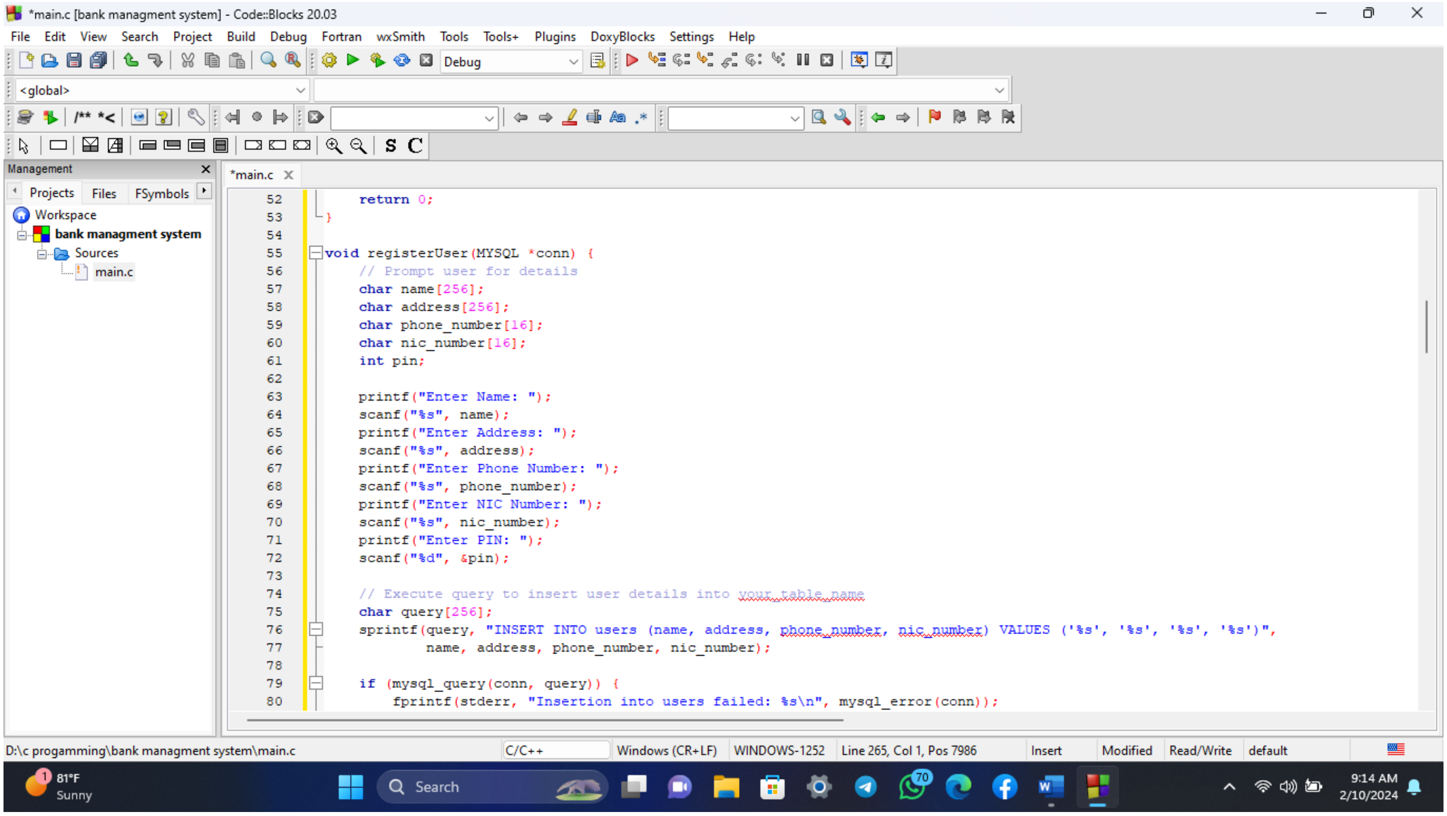
2.3.Screensots Of Code:



```c
 #include <mysql/mysql.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void registerUser(MYSQL *conn);
void loginUser(MYSQL *conn);
void checkAccountDetails(MYSQL *conn, int user_id);
void depositMoney(MYSQL *conn, int user_id);
void withdrawMoney(MYSQL *conn, int user_id);
double checkAccountBalance(MYSQL *conn, int user_id);

int main() {
    MYSQL *conn;

    conn = mysql_init(NULL);

    if (mysql_real_connect(conn, "localhost", "root", "Samitha@15", "bank_db", 0, NULL, 0)) {
        printf("Connected to MySQL server.\n");

        int choice;
        do {
            printf("\nMenu:\n");
            printf("1. Register\n");
            printf("2. Login\n");
            printf("3. Exit\n");
            printf("Enter your choice: ");
            scanf("%d", &choice);

```



```c
            scanf("%d", &choice);

            switch (choice) {
                case 1:
                    registerUser(conn);
                    printf("Returning to menu......");
                case 2:
                    loginUser(conn);
                    printf("Returning to menu......");
                case 3:
                    printf("Exiting program.\n");
                    break;
                default:
                    printf("Invalid choice. Please enter a number between 1 and 3.\n");
                    break;
            }
        } while (choice != 3);

        mysql_close(conn);
    } else {
        fprintf(stderr, "Connection failed: %s\n", mysql_error(conn));
        return 1;
    }

    return 0;
}

void registerUser(MYSQL *conn) {
    // Prompt user for details
```


*main.c [bank managment system] - Code::Blocks 20.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
Debug
<global>
Management
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Workspace
bank managment system
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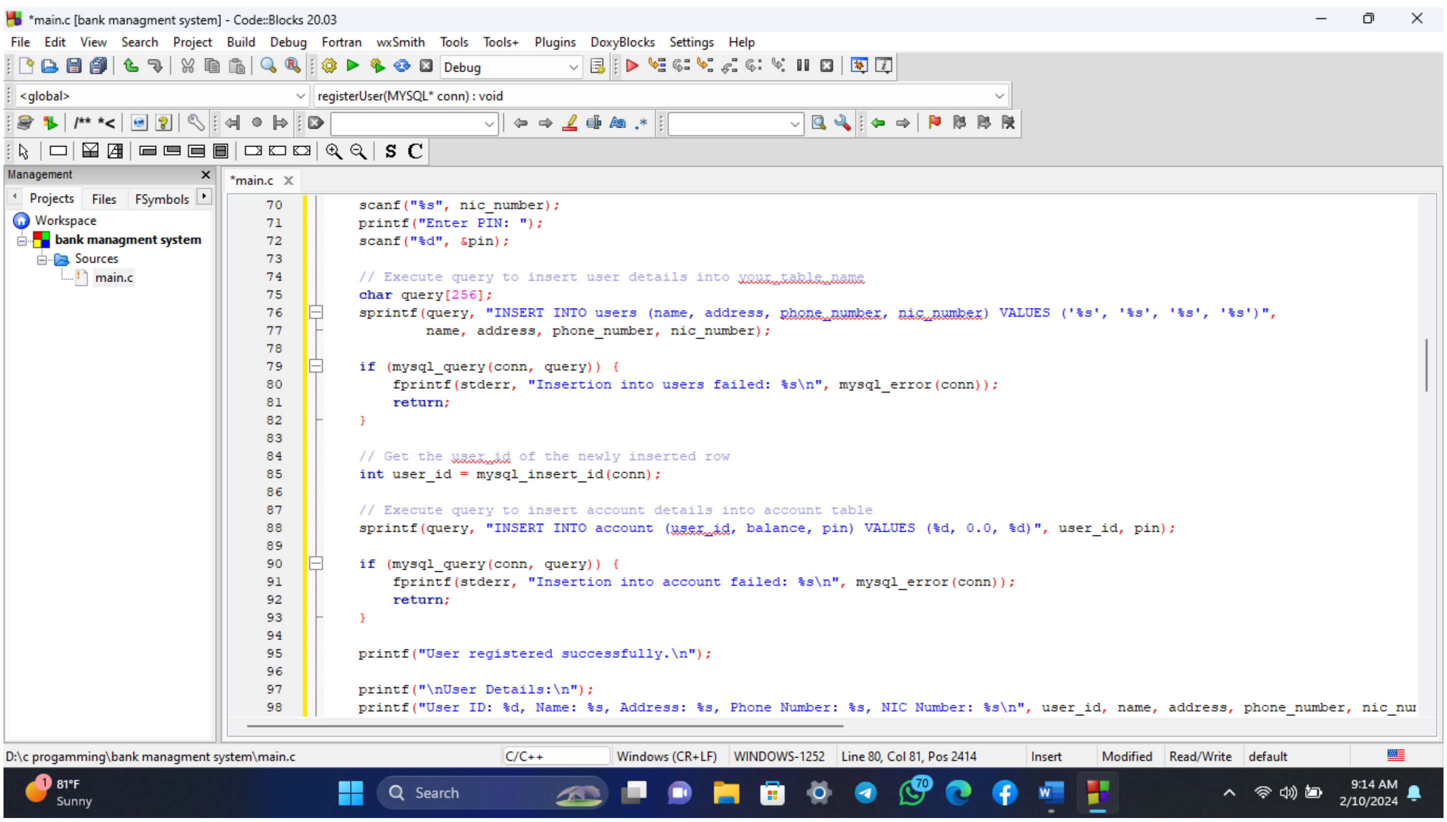
main.c
*main.c
    return 0;
}

void registerUser(MYSQL *conn) {
    // Prompt user for details
    char name[256];
    char address[256];
    char phone_number[16];
    char nic_number[16];
    int pin;

    printf("Enter Name: ");
    scanf("%s", name);
    printf("Enter Address: ");
    scanf("%s", address);
    printf("Enter Phone Number: ");
    scanf("%s", phone_number);
    printf("Enter NIC Number: ");
    scanf("%s", nic_number);
    printf("Enter PIN: ");
    scanf("%d", &pin);

    // Execute query to insert user details into your_table_name
    char query[256];
    sprintf(query, "INSERT INTO users (name, address, phone_number, nic_number) VALUES ('%s', '%s', '%s', '%s')",
            name, address, phone_number, nic_number);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Insertion into users failed: %s\n", mysql_error(conn));
D:\c progamming\bank managment system\main.c
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 265, Col 1, Pos 7986
Insert
Modified
Read/Write
default
81°F
Sunny
Search
9:14 AM
2/10/2024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<global>
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Sources
main.c
*main.c
    scanf("%s", nic_number);
    printf("Enter PIN: ");
    scanf("%d", &pin);

    // Execute query to insert user details into your_table_name
    char query[256];
    sprintf(query, "INSERT INTO users (name, address, phone_number, nic_number) VALUES ('%s', '%s', '%s', '%s')",
            name, address, phone_number, nic_number);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Insertion into users failed: %s\n", mysql_error(conn));
        return;
    }

    // Get the user_id of the newly inserted row
    int user_id = mysql_insert_id(conn);

    // Execute query to insert account details into account table
    sprintf(query, "INSERT INTO account (user_id, balance, pin) VALUES (%d, 0.0, %d)", user_id, pin);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Insertion into account failed: %s\n", mysql_error(conn));
        return;
    }

    printf("User registered successfully.\n");

    printf("\nUser Details:\n");
    printf("User ID: %d, Name: %s, Address: %s, Phone Number: %s, NIC Number: %s\n", user_id, name, address, phone_number, nic_nu
D:\c progamming\bank managment system\main.c
C/C++
Windows (CR+LF)
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Line 80, Col 81, Pos 2414
Insert
Modified
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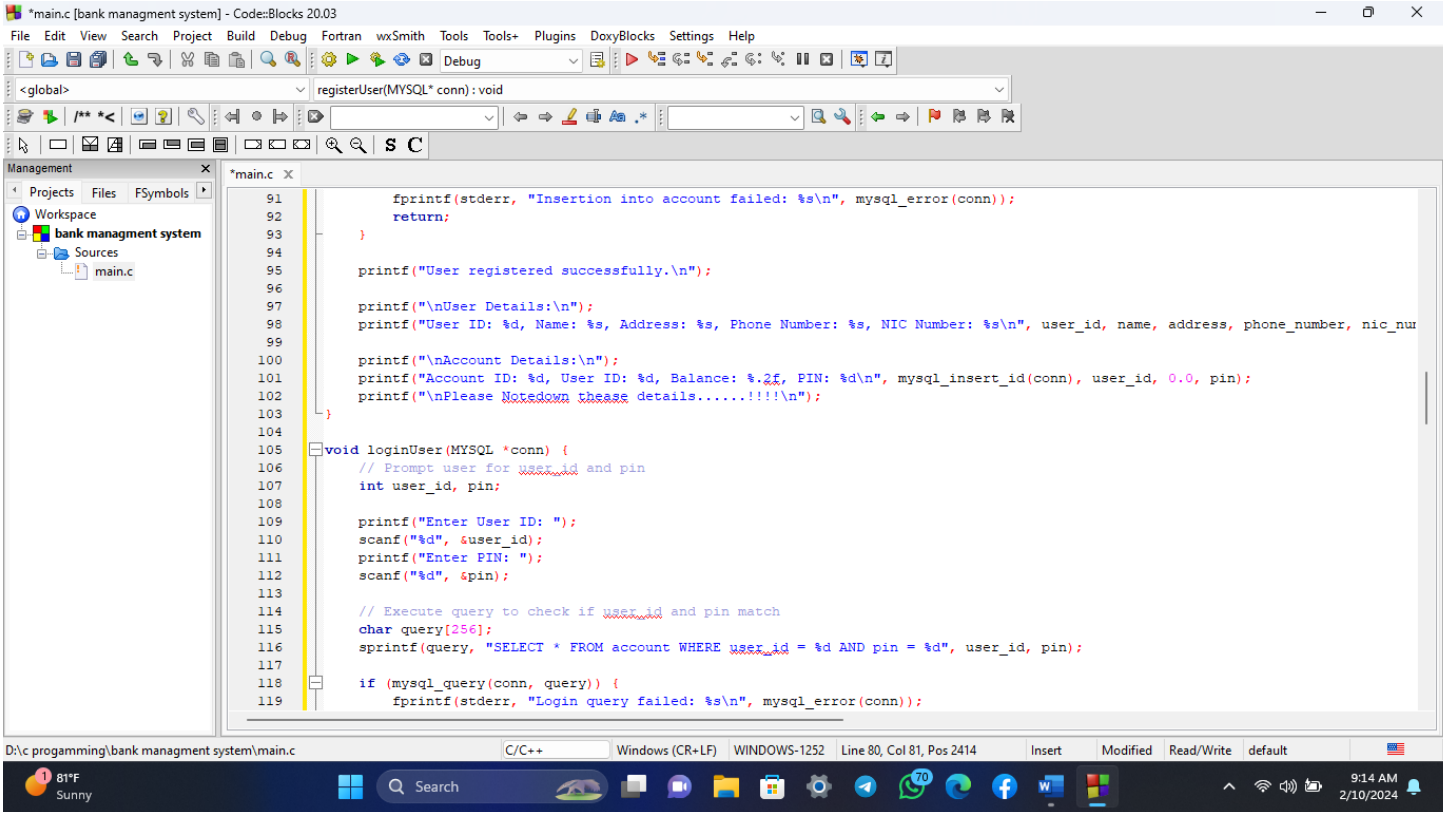
*main.c [bank managment system] - Code::Blocks 20.03
registerUser(MYSQL* conn) : void
        fprintf(stderr, "Insertion into account failed: %s\n", mysql_error(conn));
        return;
    }

    printf("User registered successfully.\n");

    printf("\nUser Details:\n");
    printf("User ID: %d, Name: %s, Address: %s, Phone Number: %s, NIC Number: %s\n", user_id, name, address, phone_number, nic_nur
    printf("\nAccount Details:\n");
    printf("Account ID: %d, User ID: %d, Balance: %.2f, PIN: %d\n", mysql_insert_id(conn), user_id, 0.0, pin);
    printf("\nPlease Notedown thease details......!!!!\n");
}

void loginUser(MYSQL *conn) {
    // Prompt user for user_id and pin
    int user_id, pin;

    printf("Enter User ID: ");
    scanf("%d", &user_id);
    printf("Enter PIN: ");
    scanf("%d", &pin);

    // Execute query to check if user_id and pin match
    char query[256];
    sprintf(query, "SELECT * FROM account WHERE user_id = %d AND pin = %d", user_id, pin);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Login query failed: %s\n", mysql_error(conn));
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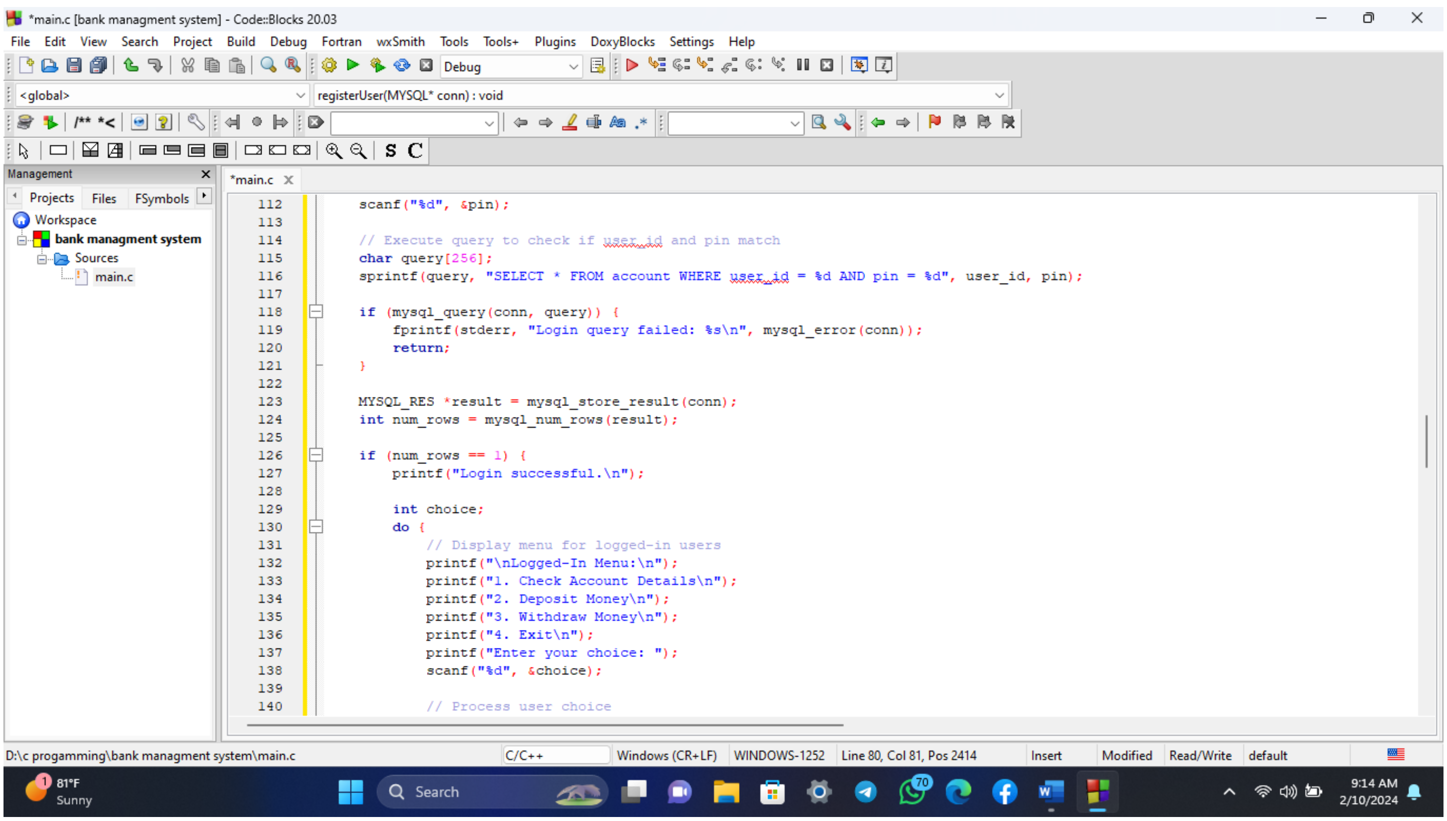
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registerUser(MYSQL* conn) : void
    scanf("%d", &pin);

    // Execute query to check if user_id and pin match
    char query[256];
    sprintf(query, "SELECT * FROM account WHERE user_id = %d AND pin = %d", user_id, pin);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Login query failed: %s\n", mysql_error(conn));
        return;
    }

    MYSQL_RES *result = mysql_store_result(conn);
    int num_rows = mysql_num_rows(result);

    if (num_rows == 1) {
        printf("Login successful.\n");

        int choice;
        do {
            // Display menu for logged-in users
            printf("\nLogged-In Menu:\n");
            printf("1. Check Account Details\n");
            printf("2. Deposit Money\n");
            printf("3. Withdraw Money\n");
            printf("4. Exit\n");
            printf("Enter your choice: ");
            scanf("%d", &choice);

            // Process user choice
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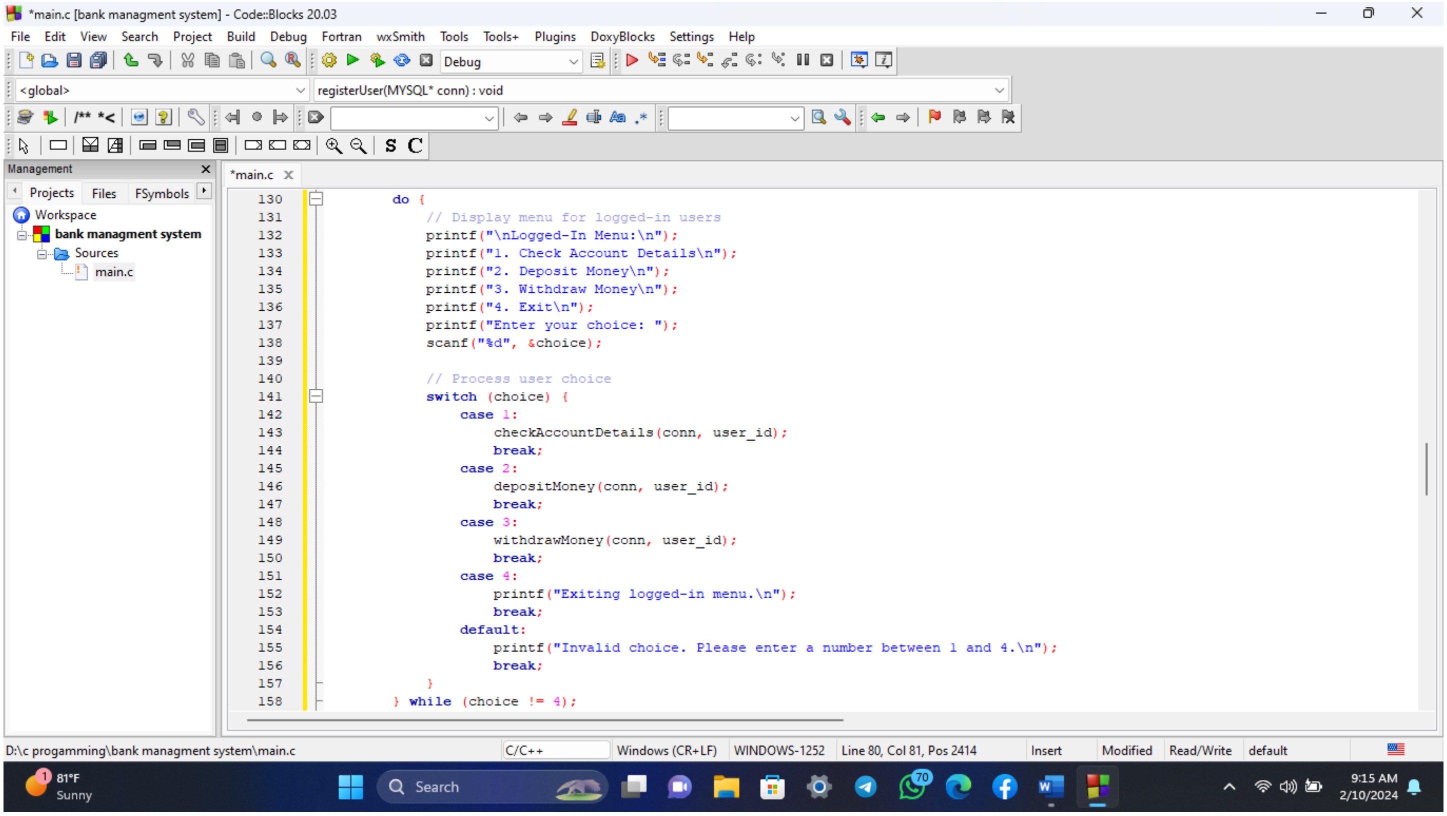
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do {
    // Display menu for logged-in users
    printf("\nLogged-In Menu:\n");
    printf("1. Check Account Details\n");
    printf("2. Deposit Money\n");
    printf("3. Withdraw Money\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    // Process user choice
    switch (choice) {
        case 1:
            checkAccountDetails(conn, user_id);
            break;
        case 2:
            depositMoney(conn, user_id);
            break;
        case 3:
            withdrawMoney(conn, user_id);
            break;
        case 4:
            printf("Exiting logged-in menu.\n");
            break;
        default:
            printf("Invalid choice. Please enter a number between 1 and 4.\n");
            break;
    }
} while (choice != 4);
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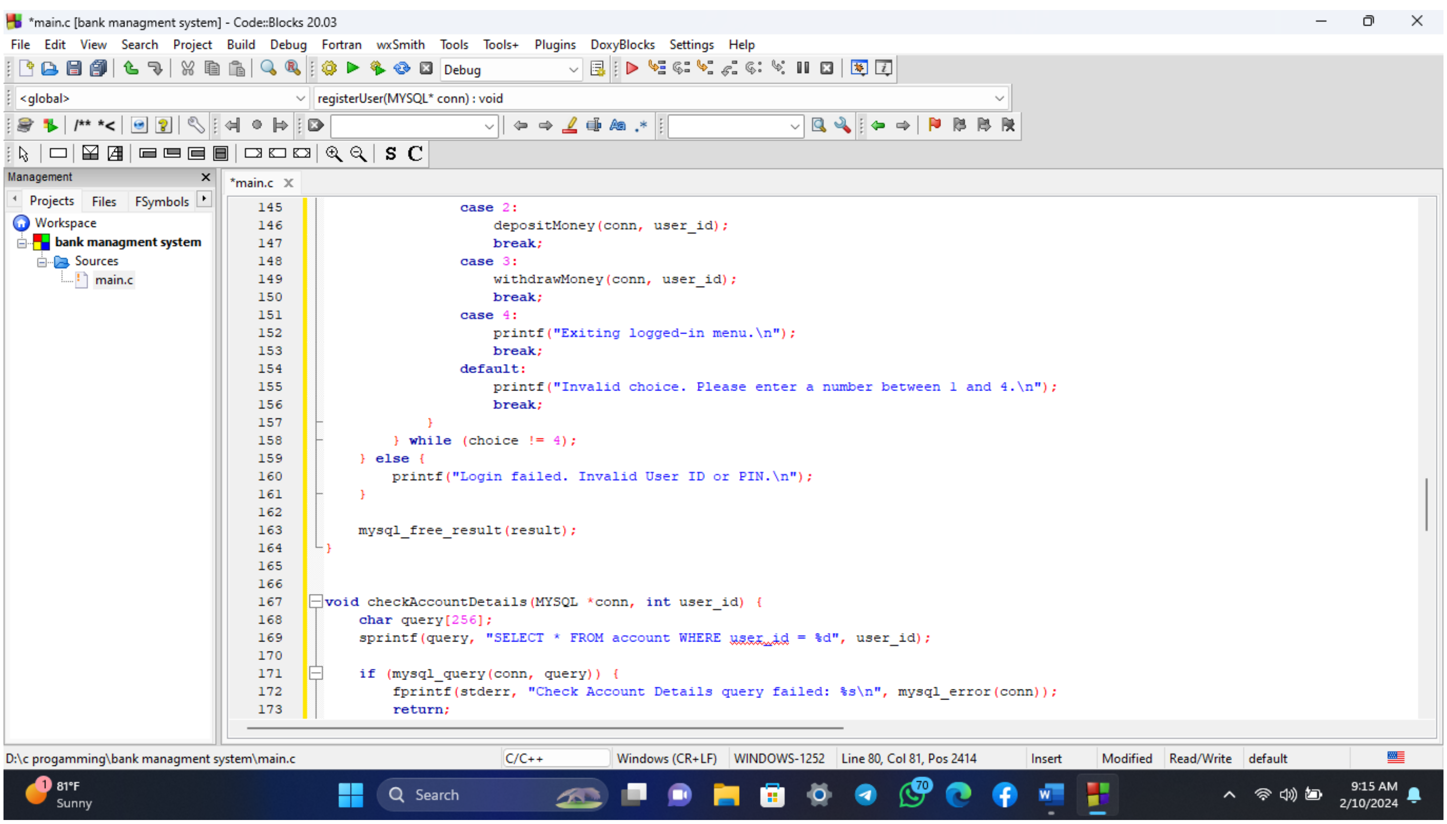
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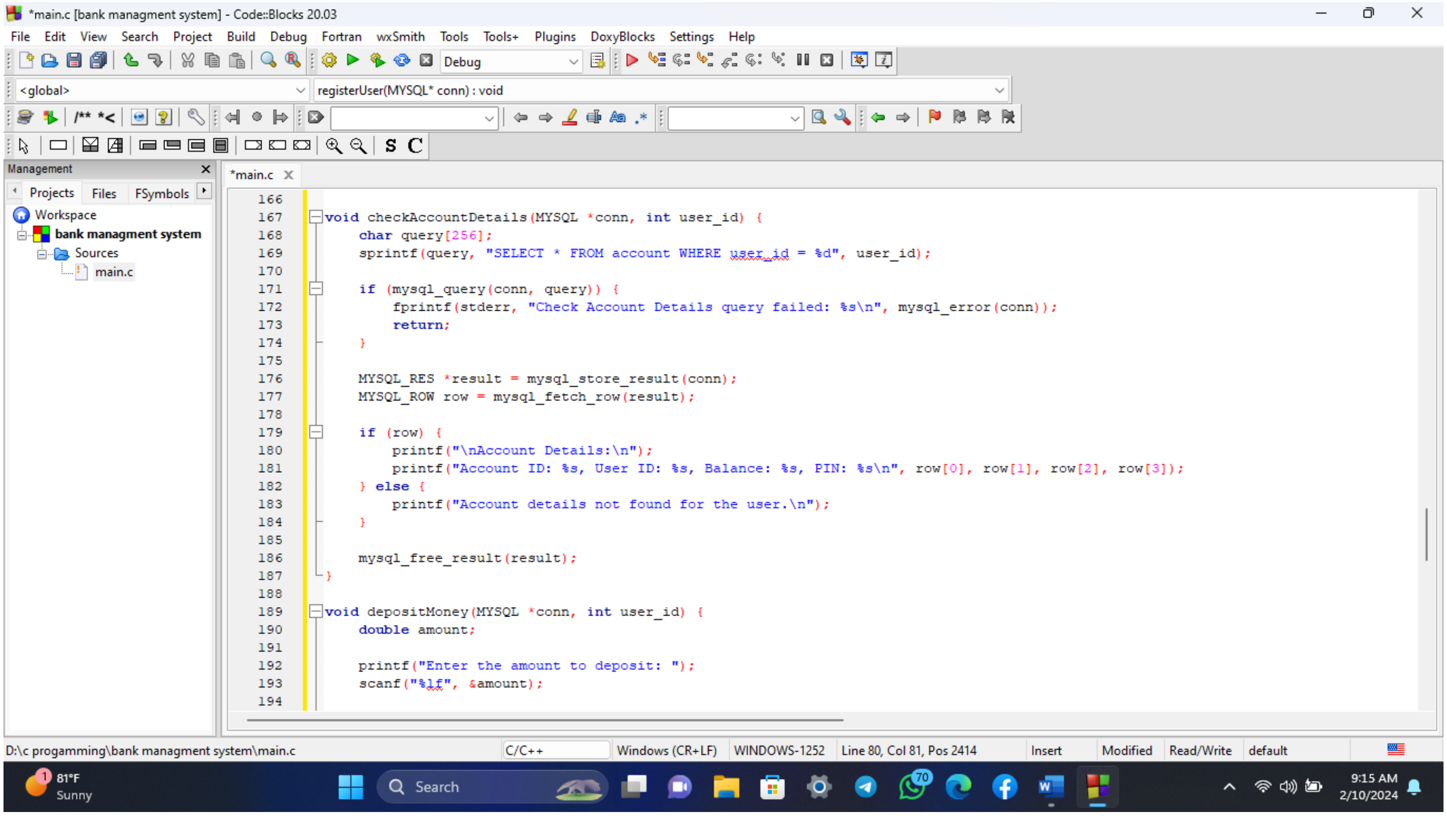
        case 2:
            depositMoney(conn, user_id);
            break;
        case 3:
            withdrawMoney(conn, user_id);
            break;
        case 4:
            printf("Exiting logged-in menu.\n");
            break;
        default:
            printf("Invalid choice. Please enter a number between 1 and 4.\n");
            break;
    }
} while (choice != 4);
} else {
    printf("Login failed. Invalid User ID or PIN.\n");
}
mysql_free_result(result);
}
void checkAccountDetails(MYSQL *conn, int user_id) {
    char query[256];
    sprintf(query, "SELECT * FROM account WHERE user_id = %d", user_id);
    if (mysql_query(conn, query)) {
        fprintf(stderr, "Check Account Details query failed: %s\n", mysql_error(conn));
        return;
D:\c progamming\bank managment system\main.c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    char query[256];
    sprintf(query, "SELECT * FROM account WHERE user_id = %d", user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Check Account Details query failed: %s\n", mysql_error(conn));
        return;
    }

    MYSQL_RES *result = mysql_store_result(conn);
    MYSQL_ROW row = mysql_fetch_row(result);

    if (row) {
        printf("\nAccount Details:\n");
        printf("Account ID: %s, User ID: %s, Balance: %s, PIN: %s\n", row[0], row[1], row[2], row[3]);
    } else {
        printf("Account details not found for the user.\n");
    }

    mysql_free_result(result);
}

void depositMoney(MYSQL *conn, int user_id) {
    double amount;

    printf("Enter the amount to deposit: ");
    scanf("%lf", &amount);

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C/C++
Windows (CR+LF)
WINDOWS-1252
Line 80, Col 81, Pos 2414
Insert
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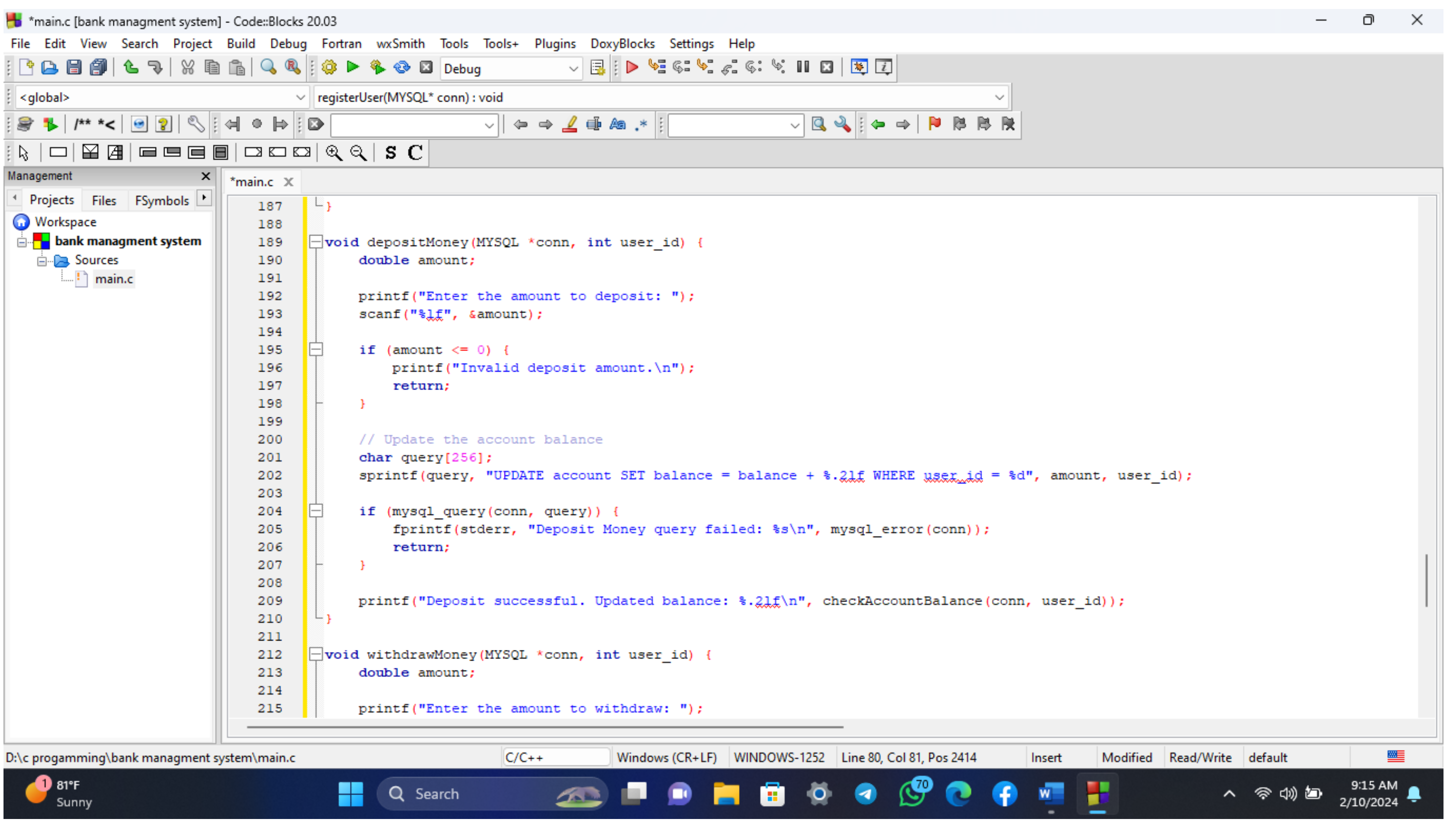
Read/Write
default
81°F
Sunny
Search
9:15 AM
2/10/2024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registerUser(MYSQL* conn) : void
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}

void depositMoney(MYSQL *conn, int user_id) {
    double amount;

    printf("Enter the amount to deposit: ");
    scanf("%lf", &amount);

    if (amount <= 0) {
        printf("Invalid deposit amount.\n");
        return;
    }

    // Update the account balance
    char query[256];
    sprintf(query, "UPDATE account SET balance = balance + %.2lf WHERE user_id = %d", amount, user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Deposit Money query failed: %s\n", mysql_error(conn));
        return;
    }

    printf("Deposit successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
}

void withdrawMoney(MYSQL *conn, int user_id) {
    double amount;

    printf("Enter the amount to withdraw: ");
D:\c progamming\bank managment system\main.c
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 80, Col 81, Pos 2414
Insert
Modified
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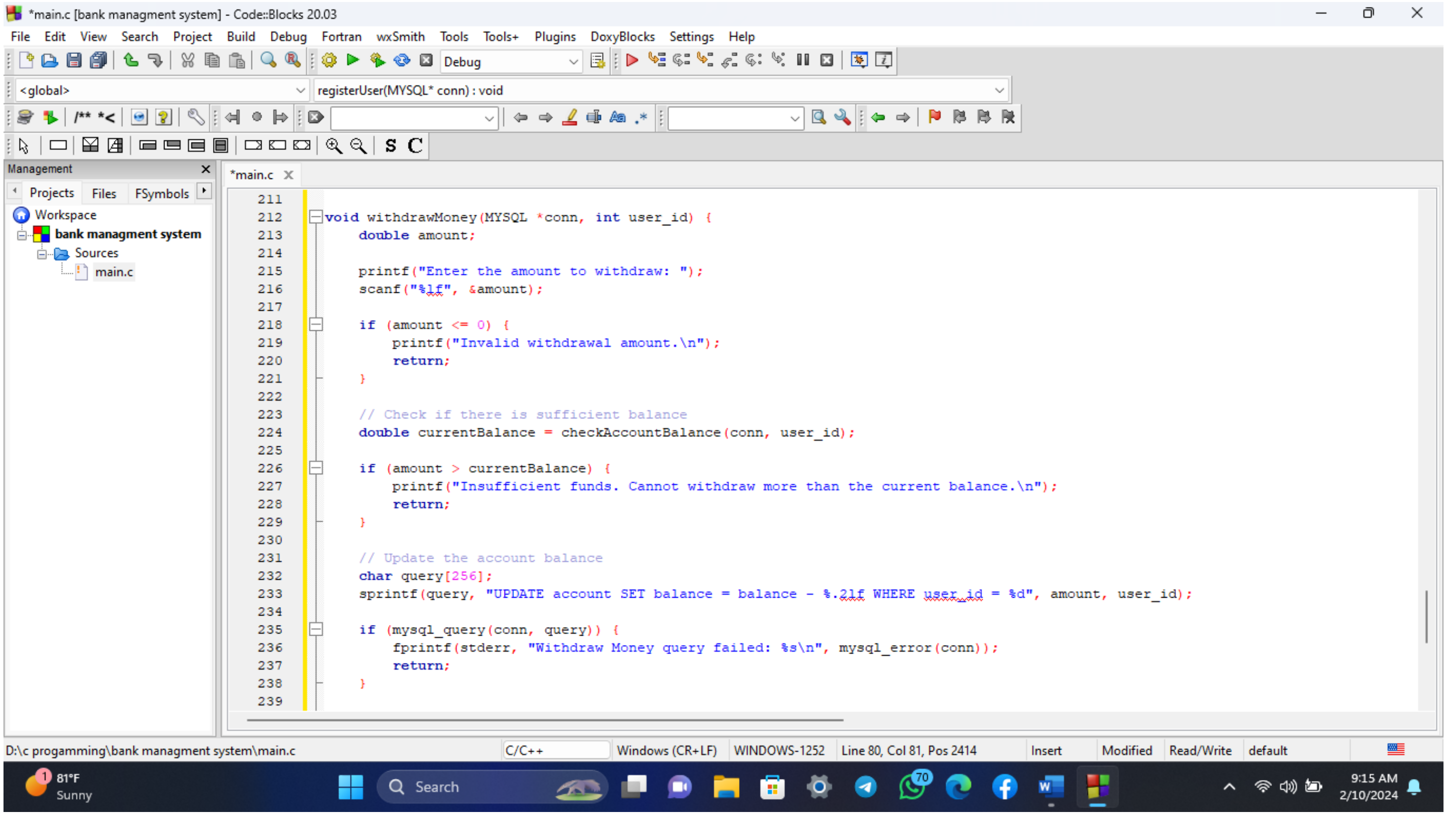
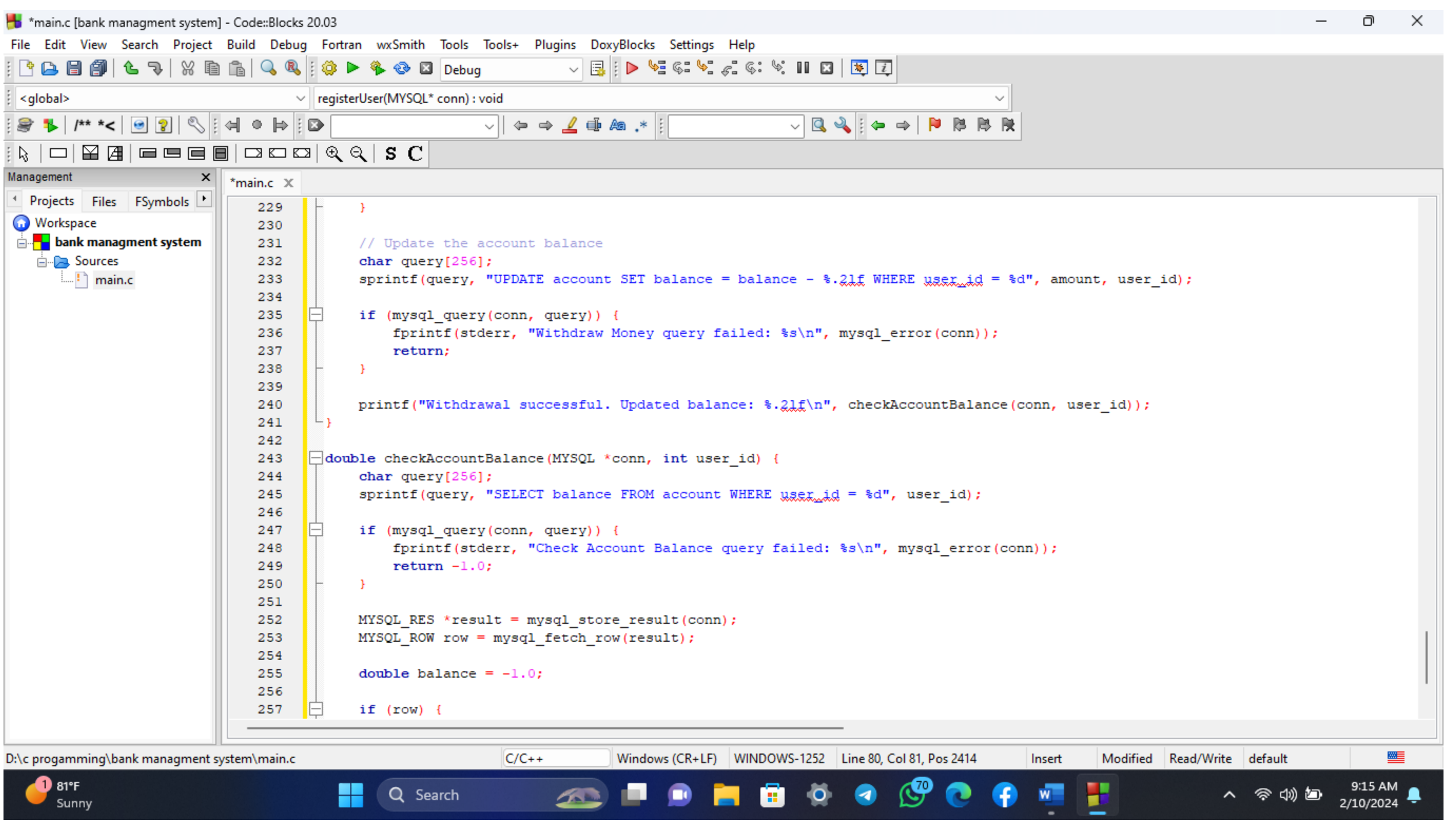
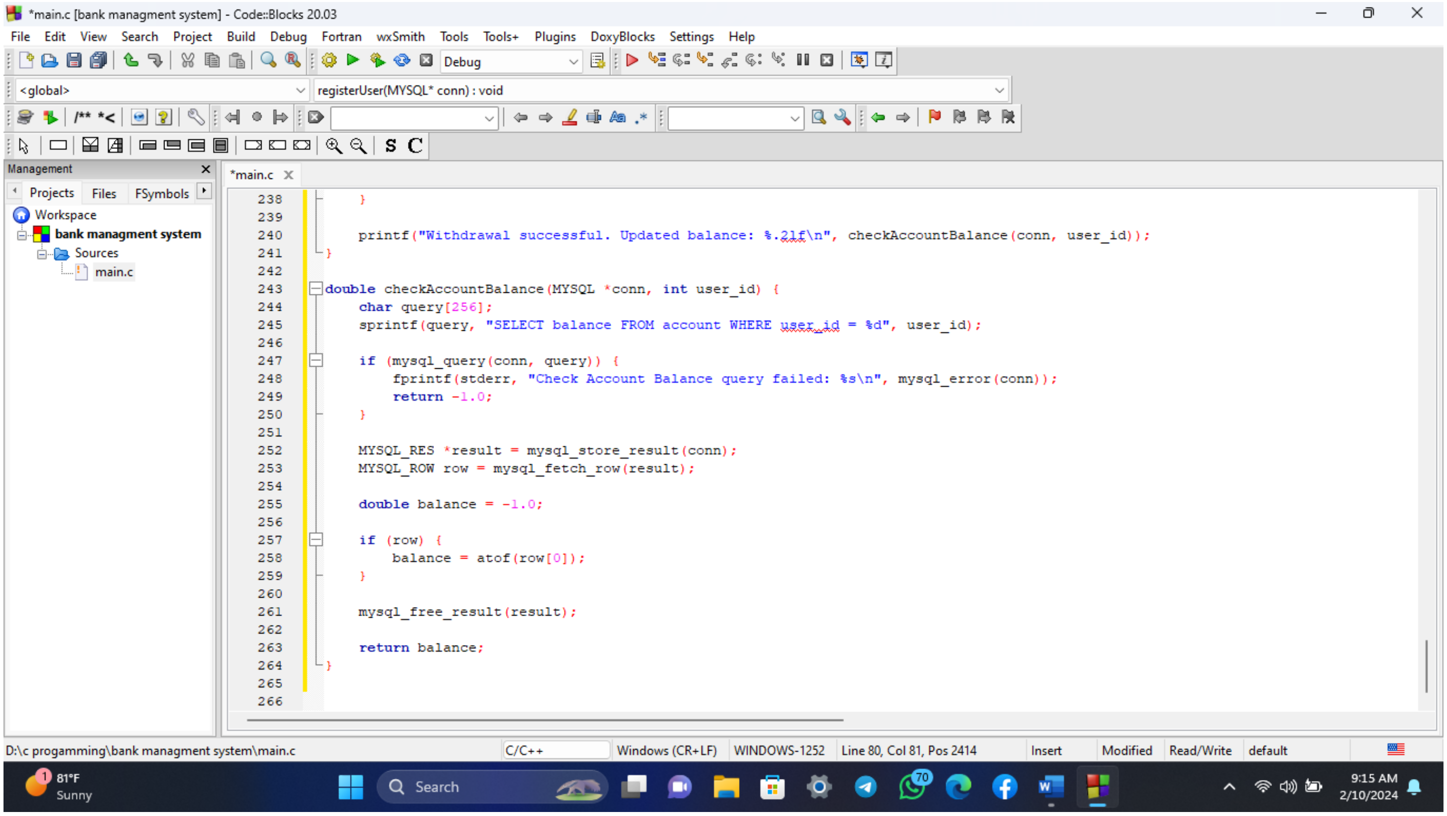


```c
void withdrawMoney(MYSQL *conn, int user_id) {
    double amount;

    printf("Enter the amount to withdraw: ");
    scanf("%lf", &amount);

    if (amount <= 0) {
        printf("Invalid withdrawal amount.\n");
        return;
    }

    // Check if there is sufficient balance
    double currentBalance = checkAccountBalance(conn, user_id);

    if (amount > currentBalance) {
        printf("Insufficient funds. Cannot withdraw more than the current balance.\n");
        return;
    }

    // Update the account balance
    char query[256];
    sprintf(query, "UPDATE account SET balance = balance - %.2lf WHERE user_id = %d", amount, user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Withdraw Money query failed: %s\n", mysql_error(conn));
        return;
    }
```

```c
    }

    // Update the account balance
    char query[256];
    sprintf(query, "UPDATE account SET balance = balance - %.2lf WHERE user_id = %d", amount, user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Withdraw Money query failed: %s\n", mysql_error(conn));
        return;
    }

    printf("Withdrawal successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
}

double checkAccountBalance(MYSQL *conn, int user_id) {
    char query[256];
    sprintf(query, "SELECT balance FROM account WHERE user_id = %d", user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Check Account Balance query failed: %s\n", mysql_error(conn));
        return -1.0;
    }

    MYSQL_RES *result = mysql_store_result(conn);
    MYSQL_ROW row = mysql_fetch_row(result);

    double balance = -1.0;

    if (row) {
```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<global>
registerUser(MYSQL* conn) : void
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Sources
main.c
*main.c
    }

    printf("Withdrawal successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
}

double checkAccountBalance(MYSQL *conn, int user_id) {
    char query[256];
    sprintf(query, "SELECT balance FROM account WHERE user_id = %d", user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Check Account Balance query failed: %s\n", mysql_error(conn));
        return -1.0;
    }

    MYSQL_RES *result = mysql_store_result(conn);
    MYSQL_ROW row = mysql_fetch_row(result);

    double balance = -1.0;

    if (row) {
        balance = atof(row[0]);
    }

    mysql_free_result(result);

    return balance;
}
D:\c progamming\bank managment system\main.c
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 80, Col 81, Pos 2414
Insert
Modified
Read/Write
default
81°F
Sunny
Search
9:15 AM
2/10/2024

2.4.Screenshots of Output:

```
tirani@tirani:~/Documents/BankingSystem$ ./main
Connected to MySQL server.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 
```

```
Menu:
1. Register
2. Login
3. Exit
Enter your choice: 1
Enter Name: tirani
Enter Address: homagama
Enter Phone Number: 0777555666
Enter NIC Number: 123456789
Enter PIN: 0000
User registered successfully.

User Details:
User ID: 4, Name: tirani, Address: homagama, Phone Number: 0777555666, NIC Number: 123456789

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Please Notedown thease details......!!!!
Returning to menu......Enter User ID: 
```

```
tirani@tirani:~/Documents/BankingSystem$ ./main
Connected to MySQL server.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 1
Enter Name: tirani
Enter Address: homagama
Enter Phone Number: 0777555666
Enter NIC Number: 123456789
Enter PIN: 0000
User registered successfully.

User Details:
User ID: 4, Name: tirani, Address: homagama, Phone Number: 0777555666, NIC Number: 123456789

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Please Notedown thease details......!!!!
Returning to menu......Enter User ID: 4
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 
```

```
tirani@tirani:~/Documents/BankingSystem$ ./main
Connected to MySQL server.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 1
Enter Name: tirani
Enter Address: homagama
Enter Phone Number: 0777555666
Enter NIC Number: 123456789
Enter PIN: 0000
User registered successfully.

User Details:
User ID: 4, Name: tirani, Address: homagama, Phone Number: 0777555666, NIC Number: 123456789

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Please Notedown thease details......!!!!
Returning to menu......Enter User ID: 4
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 1

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 2
Enter the amount to deposit: 25000
Deposit successful. Updated balance: 25000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice:
```

```
2. Login
3. Exit
Enter your choice: 1
Enter Name: tirani
Enter Address: homagama
Enter Phone Number: 0777555666
Enter NIC Number: 123456789
Enter PIN: 0000
User registered successfully.

User Details:
User ID: 4, Name: tirani, Address: homagama, Phone Number: 0777555666, NIC Number: 123456789

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Please Notedown thease details......!!!!
Returning to menu......Enter User ID: 4
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 1

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 2
Enter the amount to deposit: 25000
Deposit successful. Updated balance: 25000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 3
Enter the amount to withdraw: 5000
Withdrawal successful. Updated balance: 20000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice:
```

```
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 1

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 2
Enter the amount to deposit: 25000
Deposit successful. Updated balance: 25000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 3
Enter the amount to withdraw: 5000
Withdrawal successful. Updated balance: 20000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 4
Exiting logged-in menu.
Returning to menu......Exiting program.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 2
Enter User ID: 4
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice:
```

```
Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 2
Enter the amount to deposit: 25000
Deposit successful. Updated balance: 25000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 3
Enter the amount to withdraw: 5000
Withdrawal successful. Updated balance: 20000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 4
Exiting logged-in menu.
Returning to menu......Exiting program.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 2
Enter User ID: 4
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 1

Account Details:
Account ID: 10004, User ID: 4, Balance: 20000.00, PIN: 0

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice:
```

```
tirani@tirani:~/Documents/BankingSystem$ ./main
Connected to MySQL server.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 
```

3.Best Practices Utilized:

3.1. Code Structure:-

Designing a complete bank management system involves multiple components and functionalities. Below is a simplified example of a C program structure for a basic bank management system. This example includes structures for customer information, account details, and functions to perform common banking operations. Keep in mind that this is a basic example, and a real-world bank management system would be much more complex.

```
#include <mysql/mysql.h>

#include <stdio.h>

#include <stdlib.h>

#include <string.h>


void registerUser(MYSQL *conn);

void loginUser(MYSQL *conn);

void checkAccountDetails(MYSQL *conn, int user_id);

void depositMoney(MYSQL *conn, int user_id);

void withdrawMoney(MYSQL *conn, int user_id);

double checkAccountBalance(MYSQL *conn, int user_id);


int main() {
   MYSQL *conn;


   conn = mysql_init(NULL);


   if (mysql_real_connect(conn, "localhost", "root", "Samitha@15", "bank_db", 0, NULL, 0)) {
      printf("Connected to MySQL server.\n");


      int choice;
      do {
         printf("\nMenu:\n");
         printf("1. Register\n");
         printf("2. Login\n");
```

```
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
          case 1:
            registerUser(conn);
            printf("Returning to menu......");
          case 2:
            loginUser(conn);
            printf("Returning to menu......");
          case 3:
            printf("Exiting program.\n");
            break;
          default:
            printf("Invalid choice. Please enter a number between 1 and 3.\n");
            break;
        }
      } while (choice != 3);

      mysql_close(conn);
  } else {
      fprintf(stderr, "Connection failed: %s\n", mysql_error(conn));
      return 1;
  }

  return 0;
}

void registerUser(MYSQL *conn) {
  // Prompt user for details
```

```c
char name[256];
char address[256];
char phone_number[16];
char nic_number[16];
int pin;

printf("Enter Name: ");
scanf("%s", name);
printf("Enter Address: ");
scanf("%s", address);
printf("Enter Phone Number: ");
scanf("%s", phone_number);
printf("Enter NIC Number: ");
scanf("%s", nic_number);
printf("Enter PIN: ");
scanf("%d", &pin);

// Execute query to insert user details into your_table_name
char query[256];
sprintf(query, "INSERT INTO users (name, address, phone_number, nic_number) VALUES ('%s', '%s', '%s', '%s')",
        name, address, phone_number, nic_number);

if (mysql_query(conn, query)) {
    fprintf(stderr, "Insertion into users failed: %s\n", mysql_error(conn));
    return;
}

// Get the user_id of the newly inserted row
int user_id = mysql_insert_id(conn);

// Execute query to insert account details into account table
```

```
    sprintf(query, "INSERT INTO account (user_id, balance, pin) VALUES (%d, 0.0, %d)", user_id, pin);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Insertion into account failed: %s\n", mysql_error(conn));
        return;
    }

    printf("User registered successfully.\n");

    printf("\nUser Details:\n");
    printf("User ID: %d, Name: %s, Address: %s, Phone Number: %s, NIC Number: %s\n", user_id, name, address,
phone_number, nic_number);

    printf("\nAccount Details:\n");
    printf("Account ID: %d, User ID: %d, Balance: %.2f, PIN: %d\n", mysql_insert_id(conn), user_id, 0.0, pin);
    printf("\nPlease Notedown thease details......!!!!\n");
}

void loginUser(MYSQL *conn) {
    // Prompt user for user_id and pin
    int user_id, pin;

    printf("Enter User ID: ");
    scanf("%d", &user_id);
    printf("Enter PIN: ");
    scanf("%d", &pin);

    // Execute query to check if user_id and pin match
    char query[256];
    sprintf(query, "SELECT * FROM account WHERE user_id = %d AND pin = %d", user_id, pin);
```

```
if (mysql_query(conn, query)) {
    fprintf(stderr, "Login query failed: %s\n", mysql_error(conn));
    return;
}

MYSQL_RES *result = mysql_store_result(conn);
int num_rows = mysql_num_rows(result);

if (num_rows == 1) {
    printf("Login successful.\n");

    int choice;
    do {
        // Display menu for logged-in users
        printf("\nLogged-In Menu:\n");
        printf("1. Check Account Details\n");
        printf("2. Deposit Money\n");
        printf("3. Withdraw Money\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        // Process user choice
        switch (choice) {
            case 1:
                checkAccountDetails(conn, user_id);
                break;
            case 2:
                depositMoney(conn, user_id);
                break;
            case 3:
```

```
            withdrawMoney(conn, user_id);
            break;
          case 4:
            printf("Exiting logged-in menu.\n");
            break;
          default:
            printf("Invalid choice. Please enter a number between 1 and 4.\n");
            break;
        }
      } while (choice != 4);
    } else {
      printf("Login failed. Invalid User ID or PIN.\n");
    }

    mysql_free_result(result);
}



void checkAccountDetails(MYSQL *conn, int user_id) {
    char query[256];
    sprintf(query, "SELECT * FROM account WHERE user_id = %d", user_id);

    if (mysql_query(conn, query)) {
      fprintf(stderr, "Check Account Details query failed: %s\n", mysql_error(conn));
      return;
    }

    MYSQL_RES *result = mysql_store_result(conn);
    MYSQL_ROW row = mysql_fetch_row(result);

    if (row) {
```

```
        printf("\nAccount Details:\n");
        printf("Account ID: %s, User ID: %s, Balance: %s, PIN: %s\n", row[0], row[1], row[2], row[3]);
    } else {
        printf("Account details not found for the user.\n");
    }

    mysql_free_result(result);
}

void depositMoney(MYSQL *conn, int user_id) {
    double amount;

    printf("Enter the amount to deposit: ");
    scanf("%lf", &amount);

    if (amount <= 0) {
        printf("Invalid deposit amount.\n");
        return;
    }

    // Update the account balance
    char query[256];
    sprintf(query, "UPDATE account SET balance = balance + %.2lf WHERE user_id = %d", amount, user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Deposit Money query failed: %s\n", mysql_error(conn));
        return;
    }

    printf("Deposit successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
}
```

```
void withdrawMoney(MYSQL *conn, int user_id) {
    double amount;

    printf("Enter the amount to withdraw: ");
    scanf("%lf", &amount);

    if (amount <= 0) {
        printf("Invalid withdrawal amount.\n");
        return;
    }

    // Check if there is sufficient balance
    double currentBalance = checkAccountBalance(conn, user_id);

    if (amount > currentBalance) {
        printf("Insufficient funds. Cannot withdraw more than the current balance.\n");
        return;
    }

    // Update the account balance
    char query[256];
    sprintf(query, "UPDATE account SET balance = balance - %.2lf WHERE user_id = %d", amount, user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Withdraw Money query failed: %s\n", mysql_error(conn));
        return;
    }

    printf("Withdrawal successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
}
```

```
double checkAccountBalance(MYSQL *conn, int user_id) {
    char query[256];
    sprintf(query, "SELECT balance FROM account WHERE user_id = %d", user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Check Account Balance query failed: %s\n", mysql_error(conn));
        return -1.0;
    }

    MYSQL_RES *result = mysql_store_result(conn);
    MYSQL_ROW row = mysql_fetch_row(result);

    double balance = -1.0;

    if (row) {
        balance = atof(row[0]);
    }

    mysql_free_result(result);

    return balance;
}
```

3.2. Security Measure:-

The security measures for a bank management system are crucial to safeguard sensitive financial information and maintain the trust of customers. Here are some key security measures commonly employed in bank management systems:

1. Encryption:

All data transmitted between users and the bank's servers should be encrypted using secure protocols such as TLS (Transport Layer Security).

Stored data, including customer information and transaction records, should be encrypted to protect it from unauthorized access.

2. Access Control:

Implement strong access controls to ensure that only authorized personnel can access sensitive information.

Use multi-factor authentication (MFA) to add an extra layer of security to user accounts.

3. Firewalls and Intrusion Detection/Prevention Systems:

Deploy firewalls to monitor and control incoming and outgoing network traffic.

Use intrusion detection and prevention systems to detect and respond to potential security threats.

4. Physical Security:

Ensure physical security measures are in place to protect servers, data centers, and other critical infrastructure.

5. Security Policies and Training:

Establish and enforce security policies and procedures for employees.

Provide regular training to staff on security best practices to prevent social engineering attacks and ensure awareness of potential risks.

6. Data Backups:

Regularly backup critical data and ensure that backup systems are secure.

Test data restoration processes to ensure that information can be recovered in case of data loss or a security incident.

Implementing a combination of these security measures can help create a robust and resilient bank management system that safeguards sensitive financial information and maintains the integrity of banking operations.

4.Additional Features:

A Bank Management System typically involves a range of features to efficiently handle various banking operations. Additional features can enhance the system's functionality, security, and user experience. Here are some additional features that could be considered for a Bank Management System.

1. Biometric Authentication: Implementing biometric authentication methods such as fingerprint or facial recognition can enhance security and streamline the login process for users.

2. Multi-factor Authentication (MFA): Adding an extra layer of security by incorporating multi-factor authentication can protect user accounts from unauthorized access.

3. Mobile Banking App: Developing a mobile banking application allows customers to access their accounts, perform transactions, and manage finances on the go.

4. Personal Finance Management (PFM): Including features that help users track and manage their expenses, set financial goals, and create budgets can enhance customer satisfaction.

5. Alerts and Notifications: Providing real-time notifications for transactions, account balances, and other important events helps users stay informed and monitor their financial activities

6. Automated Customer Support: Implementing chatbots or virtual assistants for customer support can enhance user experience by providing quick responses to common queries and issues.

7. Card Management: Allowing users to manage their debit/credit cards, including setting spending limits, activating/deactivating cards, and reporting lost or stolen cards, enhances user control and security.

8. Online Loan Application and Management: Facilitating the entire loan application process online, including document submission and status tracking, can streamline the lending process.

The specific features to include would depend on the bank's size, target audience, and the regulatory environment in which it operates. Additionally, it's crucial to prioritize security and compliance when implementing new features in a Bank Management System.

Explanation:-

1. Starting from the main() function, we are declaring a variable to know which specific operation the user wants to perform into the Student Record Management Systems , but on the user's command, it will keep getting executed always. After that, we ask the user to enter the operation they want to perform.

5.Conclusion:

• In this article, we have built a mini Bank Management System in C language using the concepts of arrays.

• In this project, the user can add a new student's record, view student details, search for a specific student's record, exit and more.

• We built it by creating individual functions for each possible operation, thus improving knowledge of the code.

• We have also tried to include information deletion and information editing.

6.References:

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Workload Matrix:

Name	NSBM ID	Contribution
Ediriweera C.A.		100%
Hearath H.M.T.G.		100%
Jayasinghe K.U.		100%
Weerasinghe E.M.L.C		100%
Weerasinghe M.G.H.		100%

THANK YOU!!!!......